

Data Management Technologies

Unit 2 Consolidation Task 2

A customer data database populated with tables, tuples and fields with SQL commands. The database is designed to contain data on an organisation's customers and sales.

Define Database Structure

/ Customer Database */*

```
TABLE customer{  
  customer_id integer [primary key]  
  first_name varchar(255)  
  last_name varchar(255)  
  address_line_1 varchar(255)  
  address_line_2 varchar(255)  
  address_city varchar(14)  
  post_code varchar(9)  
  email_address varchar (255)  
  phone_number varchar (255)  
}
```

```
TABLE product{  
  product_id integer [primary key]  
  item_name varchar(255)  
  item_description varchar(255)  
  quantity_stocked integer  
  unit_price float  
}
```

```
TABLE sale_header{  
  sale_id integer [primary key]  
  customer_id integer  
  date_of_sale date  
  total_price float  
  sale_notes varchar (255)  
}
```

```
TABLE sale_detail{
  sale_detail_id varchar [primary key]
  sale_id integer // foreign key
  item_name varchar (255)
  item_description varchar (255)
  unit_price float
  quantity_ordered integer
  batch_number varchar (255)
}
```

Define Data

// Customer table

// Fields: customer id, first name, last name, address line 1, address line 2, address city, postcode, phone number, email address

001, John, Smith, 23 Boat Lane, Edgbaston, Birmingham, B23 2JH, 07756684382, johnsmith@yahoo.com

002, Mary, Baxter, 42 Ship Grove, Sheeptown, Manchester, M5 3ED, 07765784763, maryb1998@yahoo.com

003, James, Thunder, 95 Manx Lane, Tohall, York, YO12 3SD, 07743243249, james2003@gmail.com

004, Max, Grim, 32 Great Street, Lewisham, London, SE1 5AK, 08874637384, topmax@yahoo.com

// Product table

// Fields: product id, product name, item name, item description, quantity stocked, unit price

001, USB Drive, DataHold USB 50GB, 3000, 34.99

002, Computer Monitor, VP Monitors 40 inch, 100, 63.45

003, Computer Keyboard, Boshita Keyboards, 2000, 34.00

// Sale_header table

// Fields: sale id, customer id, date of sale, total price, sale notes

0001, 003, 12/02/2026, 98.44, fast delivery requested

0002, 004, 10/02/2026, 68.00, n/a

// Sale_detail table

// Fields: sale detail id, sale id, product id, item name, item description, unit price, quantity ordered, batch number

01-0001, 0001, 001, USB Drive, Datasafe USB 50GB, 34.99, 3, Batch 3904N

02-0001, 0001, 002, Computer Monitor, VP Monitors 40 inch, 63.45, 1, Batch 3423M

01-0002, 0002, 003, Computer Keyboard, Boshita Keyboards, 34.00, 1, Batch 344fn

Creating SQL Commands

// Command to print the whole sale detail table

SELECT * FROM sale_detail

// Command to print sale detail data for sale ID 0001

SELECT * FROM sale_detail **where** (sale_id = 0001)

// Command to print all customer details

SELECT * FROM customer